KUDZAI NEVANJI

AI / Machine Learning Engineer • IoT Security Researcher

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PROFESSIONAL SUMMARY

Master's graduate in Computer Engineering specializing in machine learning and IoT security, with hands-on experience building and deploying deep-learning systems on Linux and Docker-based infrastructure. Author of a published IEEE Access paper and a first-author manuscript under major revision. Skilled in Python, TensorFlow/PyTorch, and end-to-end model development — from data collection and anomaly detection to policy-driven deployment. A self-motivated engineer who thrives under deadlines, adapts quickly to new environments, and works effectively across multicultural teams.

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript (Node.js), SQL

ML / AI: TensorFlow, PyTorch, LSTM / Autoencoders, NLP, Computer Vision, Deep Learning, Anomaly Detection

Backend / APIs: FastAPI, REST APIs, Node.js

Data / Infra: Linux, Docker, Git, PostgreSQL / relational databases, Alluxio, Ceph

IoT / Security: Matter, Smart-Home Protocols, Embedded Security, Access Control (ACL / MFA)

LLM Tooling: Prompt engineering & application development with ChatGPT and Claude

EDUCATION

M.S. in Computer Engineering *Sep 2024 – Aug 2026*

Ajou University — Suwon, South Korea

- Research focus: Behaviour-Aware Policy Engine for Matter smart homes — anomaly detection and policy enforcement for secure IoT access control.

B.S. in Business Studies & Computing Science *Feb 2018 – Feb 2022*

University of Zimbabwe

- Upper Second Class Honours (2.1).

EXPERIENCE

Graduate Research Assistant *Sep 2024 – Aug 2026*

DBDC Lab, Ajou University

- Designed and implemented a behaviour-aware security framework for Matter smart-home hubs using an LSTM Autoencoder to model normal activity and detect anomalous commands.
- Built a Behavioural Intent Verification module that converts reconstruction errors into deviation scores, mapped to graded enforcement actions (Accept / Confirm / MFA / Block).
- Developed and trained models in Python (TensorFlow/PyTorch), deploying and evaluating them on Linux servers and Docker-based systems.
- Authored a first-author manuscript under major revision at IEEE Access (2026) and co-authored a published IEEE Access paper on cross-layer caching.

IT Attaché *Jul 2020 – Jun 2021*

Chitungwiza Municipality — Zimbabwe

- Delivered end-to-end IT support: hardware/software setup, network and internet troubleshooting, and staff technical support.
- Managed the organisation's website (WordPress) and installed/configured database systems (SAGE Evolution).

SELECTED PROJECTS

LSTM-AE Behavioural Intent Verification for Matter Hubs *2024 – 2026*

A behaviour-aware security layer for Matter smart-home hubs. Where traditional Matter relies only on authentication and ACL authorization, this system learns normal activity patterns from device event logs via an LSTM Autoencoder, scores behavioural deviations, and enforces graded policy responses through a Behaviour-Aware Policy Engine.

Stack: Python, TensorFlow/PyTorch, LSTM Autoencoders, Docker, Linux

Cross-Layer Caching for High-Performance Computing *2025 – 2026*

Experimental evaluation of Alluxio and Ceph tiering on HACC-IO workloads, demonstrating that hybrid architectures bridge read efficiency and write throughput through intelligent data placement across layers.

Stack: Alluxio, Ceph, FUSE, HACC-IO, Linux

PUBLICATIONS

K. Nevanji, P. Hamandawana, T.-S. Chung, "Behaviour-Aware Access Control for Matter Hubs via LSTM-AE-Based Deviation Scoring," *IEEE Access*, 2026. *(Major revision – resubmitted)*

B. J. Siwonde, P. Hamandawana, **K. Nevanji**, T.-S. Chung, "Cross-Layer Caching for High-Performance Computing: Experimental Analysis of Alluxio, Ceph Tiering, and Hybrid Designs Using HACC-IO Traces," *IEEE Access*, Feb. 2026. doi.org/10.1109/ACCESS.2026.3664402

LANGUAGES & ADDITIONAL

Languages: English (fluent), Korean (intermediate), Sign Language

Interests: Travel, reading, and providing technical support for friends and family

REFERENCES

Professor Tae-Sun Chung — Department of Software Engineering, Ajou University
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